

Unit B Review
Multiple Choice Bank — 75 Questions

Question Set

MC 1. Which cellular mechanism explains the obligatory partnership between theca cells and granulosa cells in estrogen production?

- A. Theca cells have aromatase enzymes but lack androgen synthesis pathways
- B. Granulosa cells produce androgens that theca cells convert to estrogen
- C. Theca cells produce androgens but lack aromatase to convert them to estrogen
- D. Both cell types independently produce estrogen through separate pathways

Correct answer: C

Rationale: Theca cells contain the complete steroidogenic pathway to convert cholesterol to androgens but lack aromatase enzymes. Granulosa cells possess aromatase to convert these androgens to estrogen, creating the two-cell, two-gonadotropin mechanism.

MC 2. During follicular development, what triggers the transformation of a primary follicle into a secondary follicle?

- A. LH surge causing ovulation
- B. FSH stimulation causing granulosa cell proliferation
- C. Progesterone from the corpus luteum
- D. Estrogen negative feedback on the pituitary

Correct answer: B

Rationale: FSH stimulation causes granulosa cells to proliferate and form fluid-filled spaces, transforming the primary follicle into a secondary follicle with increased estrogen production capacity.

MC 3. What structural feature of the Fallopian tubes specifically facilitates capture of the ovulated oocyte?

- A. Smooth muscle layers creating peristaltic movements
- B. Secretory cells producing nutrients
- C. Fimbriae with coordinated ciliary beating
- D. The wider diameter of the ampulla region

Correct answer: C

Rationale: The fimbriae at the tube opening use coordinated ciliary beating to capture the released ovum during ovulation from the peritoneal cavity.

MC 4. In controlled ovarian stimulation for assisted reproductive technologies, why do fertility specialists monitor estradiol levels?

- A. To assess follicular maturity and estrogen production capacity
- B. To determine progesterone production by the corpus luteum
- C. To measure androgen conversion efficiency
- D. To evaluate endometrial thickness directly

Correct answer: A

Rationale: Ultrasound tracking of follicular size correlates with estrogen production capacity, and monitoring estradiol levels helps assess follicular maturity for precise timing of oocyte retrieval.

MC 5. Which molecular mechanism explains endometrial thickening during the proliferative phase?

- A. Progesterone receptor activation upregulating glycogen accumulation genes
- B. Estrogen receptor activation upregulating DNA synthesis and cell division genes
- C. LH receptor activation stimulating vascular maturation

D. FSH receptor activation promoting secretory protein production

Correct answer: B

Rationale: Estrogen receptor activation during the proliferative phase upregulates genes for DNA synthesis, cell division, and angiogenesis, causing endometrial thickening.

MC 6. What characterizes the Graafian stage of follicular development?

- A. Primary oocyte beginning meiosis II
- B. Formation of the corpus luteum
- C. Large antrum filled with follicular fluid containing hormones and growth factors
- D. Initial response to FSH stimulation

Correct answer: C

Rationale: The dominant follicle reaches Graafian stage with a large antrum filled with follicular fluid containing hormones and growth factors before ovulation.

MC 7. During the LH surge, what immediate structural change occurs to the dominant follicle?

- A. Granulosa cells begin proliferating rapidly
- B. Enzymatic breakdown of the follicle wall
- C. Formation of fluid-filled spaces
- D. Theca cells start producing androgens

Correct answer: B

Rationale: During ovulation triggered by the LH surge, enzymatic breakdown of the follicle wall releases the secondary oocyte surrounded by cumulus cells.

MC 8. Which cells in the ovarian follicle possess FSH receptors and aromatase enzymes?

- A. Theca cells
- B. Granulosa cells
- C. Cumulus cells
- D. Primary oocytes

Correct answer: B

Rationale: Granulosa cells possess FSH receptors and aromatase enzymes but lack the enzymatic machinery to produce androgens from cholesterol.

MC 9. What is the functional significance of the ampulla region of the Fallopian tube?

- A. It produces the LH surge for ovulation
- B. It provides the optimal environment for fertilization
- C. It secretes progesterone to maintain pregnancy
- D. It contains the fimbriae for oocyte capture

Correct answer: B

Rationale: The tube's ampulla region provides the optimal environment for fertilization through its wider diameter and specialized secretory cells that produce nutrients for sperm capacitation and early embryonic development.

MC 10. What cellular cooperation enables the two-cell, two-gonadotropin mechanism of hormone production?

- A. Theca cells respond to FSH while granulosa cells respond to LH
- B. Both cell types produce complete steroid hormones independently
- C. Theca-produced androgens diffuse to granulosa cells for conversion to estrogen
- D. Granulosa cells produce progesterone that theca cells convert to estrogen

Correct answer: C

Rationale: Theca-produced androgens diffuse across the basement membrane to granulosa cells for conversion to estrogen through the two-cell, two-gonadotropin mechanism.

MC 11. During the secretory phase, what gene expression changes does progesterone receptor activation induce?

- A. Upregulation of DNA synthesis and cell division genes
- B. Shift toward secretory protein production and glycogen accumulation
- C. Increased androgen synthesis pathways
- D. Enhanced FSH receptor expression

Correct answer: B

Rationale: Progesterone receptor activation during the secretory phase shifts gene expression toward secretory protein production, glycogen accumulation, and vascular maturation to support potential implantation.

MC 12. What happens to the follicle remnant after ovulation during the LH surge?

- A. It degenerates immediately into atretic tissue
- B. It becomes a primary follicle for the next cycle
- C. It transforms into a corpus luteum through luteinization
- D. It releases additional oocytes

Correct answer: C

Rationale: The LH surge transforms the follicle remnant into a corpus luteum through luteinization, which then produces progesterone.

MC 13. Which specialized cells create peristaltic movements in the Fallopian tubes?

- A. Ciliated epithelial cells
- B. Smooth muscle layers
- C. Secretory cells
- D. Fimbrial cells

Correct answer: B

Rationale: The Fallopian tubes feature smooth muscle layers that create coordinated peristaltic movements to transport the ovulated oocyte toward the uterus.

MC 14. In a primordial follicle, what is the meiotic status of the oocyte?

- A. Completed meiosis II
- B. Arrested in meiosis II
- C. Arrested in meiosis I
- D. Not yet begun meiosis

Correct answer: C

Rationale: Each follicle contains a primary oocyte arrested in meiosis I, surrounded by granulosa cells that provide nutrients and respond to hormonal signals.

MC 15. What triggers the decidual reaction in the endometrium?

- A. FSH stimulation during the follicular phase
- B. Estrogen peak at ovulation
- C. Progesterone from corpus luteum if implantation occurs
- D. LH surge during ovulation

Correct answer: C

Rationale: The corpus luteum produces progesterone that maintains the secretory endometrium and creates a decidual reaction if implantation occurs.

MC 16. Which cells surrounding the granulosa layer produce androgens in the follicle?

- A. Cumulus cells
- B. Theca cells
- C. Primary oocytes
- D. Luteal cells

Correct answer: B

Rationale: The theca cells surrounding the granulosa layer produce androgens that granulosa cells convert to estrogen through aromatase enzyme activity.

MC 17. What is the primary function of cumulus cells released with the secondary oocyte?

- A. Produce progesterone for implantation
- B. Surround and support the oocyte during ovulation
- C. Generate the LH surge
- D. Form the corpus luteum

Correct answer: B

Rationale: During ovulation, the secondary oocyte is released surrounded by cumulus cells into the peritoneal cavity.

MC 18. In assisted reproductive technologies, what substance is used to mimic the LH surge?

- A. Exogenous FSH
- B. Synthetic estrogen
- C. hCG (human chorionic gonadotropin)
- D. Progesterone supplements

Correct answer: C

Rationale: Fertility specialists trigger ovulation with hCG that mimics the LH surge for precise timing of procedures.

MC 19. What structural feature allows precise hormonal communication between theca and granulosa cells?

- A. Direct cytoplasmic connections
- B. Androgens diffusing across the basement membrane
- C. Shared blood supply
- D. Neural connections

Correct answer: B

Rationale: Theca-produced androgens diffuse across the basement membrane to granulosa cells for conversion to estrogen.

MC 20. During the follicular phase, what effect do rising estrogen levels have on cervical mucus?

- A. Makes it hostile to sperm
- B. Causes changes that facilitate sperm transport
- C. Triggers mucus production to stop
- D. Increases acidity to prevent infection

Correct answer: B

Rationale: Increasing estrogen levels cause cervical mucus changes that facilitate sperm transport during the follicular phase.

MC 21. What enzymatic activity distinguishes granulosa cells from theca cells?

- A. Ability to produce androgens from cholesterol
- B. Presence of aromatase to convert androgens to estrogen
- C. Capacity to synthesize progesterone
- D. Expression of LH receptors

Correct answer: B

Rationale: Granulosa cells possess aromatase enzymes to convert androgens to estrogen, while theca cells lack this enzyme despite having the complete steroidogenic pathway.

MC 22. Which hormone receptors do theca cells contain that enable their steroidogenic function?

- A. FSH receptors
- B. Estrogen receptors
- C. LH receptors
- D. Progesterone receptors

Correct answer: C

Rationale: Theca cells contain LH receptors and the complete steroidogenic pathway to convert cholesterol to androgens.

MC 23. What specialized secretory function do cells in the ampulla provide?

- A. Hormones to trigger ovulation
- B. Nutrients for sperm capacitation and early embryonic development
- C. Enzymes to break down the follicle wall
- D. Mucus to prevent bacterial infection

Correct answer: B

Rationale: Specialized secretory cells in the ampulla produce nutrients for sperm capacitation and early embryonic development.

MC 24. What is contained within the follicular fluid of a Graafian follicle?

- A. Only estrogen
- B. Primary oocytes
- C. Hormones and growth factors
- D. Sperm cells

Correct answer: C

Rationale: The Graafian follicle contains a large antrum filled with follicular fluid containing hormones and growth factors.

MC 25. During endometrial proliferation, which transcription factor activity is primarily responsible?

- A. Activated progesterone receptors
- B. Activated estrogen receptors
- C. FSH-receptor complexes
- D. LH-receptor complexes

Correct answer: B

Rationale: Estrogen receptor activation acts as a transcription factor during the proliferative phase, upregulating genes for DNA synthesis, cell division, and angiogenesis.

MC 26. What happens to granulosa cells when FSH levels rise during the follicular phase?

- A. They undergo apoptosis
- B. They proliferate and increase estrogen production
- C. They convert to theca cells
- D. They stop producing aromatase

Correct answer: B

Rationale: Rising FSH levels stimulate granulosa cell proliferation and estrogen production in developing follicles.

MC 27. Which structural arrangement enables the ovaries to function as both endocrine glands and gamete-producing organs?

- A. Separation of follicles from blood vessels
- B. Specialized follicular structures with layers of supportive cells
- C. Direct neural connections to the pituitary
- D. Continuous production of new oocytes

Correct answer: B

Rationale: The ovaries function through specialized follicular structures that house developing oocytes within layers of supportive cells that also produce hormones.

MC 28. What cellular process characterizes luteinization after ovulation?

- A. Granulosa cells begin producing androgens
- B. The follicle remnant transforms to produce progesterone
- C. New oocytes begin developing
- D. Theca cells lose their LH receptors

Correct answer: B

Rationale: The LH surge transforms the follicle remnant into a corpus luteum through luteinization, which produces progesterone.

MC 29. In the context of fertility treatment, why is ultrasound tracking of follicular size important?

- A. It directly measures progesterone levels
- B. It correlates with estrogen production capacity
- C. It shows endometrial thickness
- D. It identifies corpus luteum formation

Correct answer: B

Rationale: Ultrasound tracking of follicular size correlates with estrogen production capacity, allowing precise timing of oocyte retrieval.

MC 30. What mechanism allows the fimbriae to capture the ovulated oocyte effectively?

- A. Suction created by muscle contractions
- B. Chemical attraction to the oocyte
- C. Coordinated ciliary beating
- D. Hormonal signaling from the ovary

Correct answer: C

Rationale: The fimbriae at the tube opening use coordinated ciliary beating to capture the released ovum during ovulation.

MC 31. During controlled ovarian stimulation, what is the purpose of using exogenous FSH?

- A. To trigger the LH surge
- B. To stimulate multiple follicle development
- C. To maintain the corpus luteum
- D. To prevent ovulation

Correct answer: B

Rationale: Fertility specialists use exogenous FSH to stimulate multiple follicle development in assisted reproductive technologies.

MC 32. What prevents granulosa cells from producing androgens independently?

- A. Presence of aromatase enzymes
- B. Lack of LH receptors

- C. Lack of enzymatic machinery to produce androgens from cholesterol
- D. Excessive estrogen production

Correct answer: C

Rationale: Granulosa cells lack the enzymatic machinery to produce androgens from cholesterol, creating dependency on theca cells.

MC 33. Which gene expression pattern characterizes the secretory phase endometrium?

- A. Upregulated DNA synthesis genes
- B. Increased cell division genes
- C. Enhanced secretory protein production and glycogen accumulation
- D. Reduced vascular development

Correct answer: C

Rationale: Progesterone receptor activation shifts gene expression toward secretory protein production, glycogen accumulation, and vascular maturation.

MC 34. What structural feature of Fallopian tubes enables nutrient provision for early embryonic development?

- A. Ciliated epithelial cells
- B. Specialized secretory cells in the ampulla
- C. Smooth muscle contractions
- D. Fimbrial movements

Correct answer: B

Rationale: The ampulla's specialized secretory cells produce nutrients for sperm capacitation and early embryonic development.

MC 35. During follicular development, what causes the formation of fluid-filled spaces?

- A. LH surge effects
- B. Progesterone secretion
- C. FSH-stimulated granulosa cell proliferation
- D. Corpus luteum formation

Correct answer: C

Rationale: FSH stimulation causes granulosa cells to proliferate and form fluid-filled spaces during follicular maturation.

MC 36. What is the functional significance of peak estrogen levels in the menstrual cycle?

- A. Inhibits FSH production permanently
- B. Triggers the LH surge leading to ovulation
- C. Causes corpus luteum degeneration
- D. Prevents follicular development

Correct answer: B

Rationale: The LH surge triggered by peak estrogen levels causes ovulation and corpus luteum formation.

MC 37. Which cells provide nutrients to the primary oocyte within the follicle?

- A. Theca cells
- B. Granulosa cells
- C. Luteal cells
- D. Epithelial cells

Correct answer: B

Rationale: Granulosa cells surround the primary oocyte and provide nutrients while responding to hormonal signals.

MC 38. What type of movement do ciliated epithelial cells create in the Fallopian tubes?

- A. Peristaltic contractions
- B. Transport assistance through ciliary beating
- C. Enzymatic breakdown
- D. Hormonal secretion

Correct answer: B

Rationale: Ciliated epithelial cells assist in oocyte transport through coordinated beating, while smooth muscle creates peristaltic movements.

MC 39. In the two-cell, two-gonadotropin mechanism, which gonadotropin primarily stimulates theca cells?

- A. FSH
- B. LH
- C. hCG
- D. GnRH

Correct answer: B

Rationale: Theca cells contain LH receptors, making them responsive to LH in the two-cell, two-gonadotropin mechanism.

MC 40. What happens to the endometrium when progesterone maintains it during the secretory phase?

- A. Rapid cell division and thinning
- B. Vascular maturation and preparation for implantation
- C. Complete shedding
- D. Cessation of all metabolic activity

Correct answer: B

Rationale: Progesterone maintains the secretory endometrium with vascular maturation to support potential implantation.

MC 41. Which enzyme activity is unique to granulosa cells in the follicle?

- A. Cholesterol synthesis
- B. Androgen production
- C. Aromatase activity
- D. Progesterone synthesis

Correct answer: C

Rationale: Granulosa cells uniquely possess aromatase enzymes to convert theca-derived androgens to estrogen.

MC 42. During ovulation, where is the secondary oocyte initially released?

- A. Directly into the Fallopian tube
- B. Into the uterus
- C. Into the peritoneal cavity
- D. Into the bloodstream

Correct answer: C

Rationale: During ovulation, the secondary oocyte surrounded by cumulus cells is released into the peritoneal cavity.

MC 43. What molecular mechanism enables estrogen and progesterone to regulate gene expression?

- A. Direct enzyme activation
- B. Binding to nuclear receptors that act as transcription factors
- C. Cell surface receptor activation only
- D. Protein synthesis inhibition

Correct answer: B

Rationale: Estrogen and progesterone bind to nuclear receptors that act as transcription factors to regulate gene expression.

MC 44. In assisted reproductive technologies, what measurement helps determine oocyte retrieval timing?

- A. Progesterone levels only
- B. Endometrial thickness alone
- C. Follicular size and estradiol levels
- D. LH concentration exclusively

Correct answer: C

Rationale: Monitoring estradiol levels to assess follicular maturity and ultrasound tracking of follicular size allows precise timing of oocyte retrieval.

MC 45. What characterizes the relationship between follicular cell types in hormone production?

- A. Independent hormone production by each cell type
- B. Obligatory partnership for estrogen synthesis
- C. Competition for hormone precursors
- D. Random hormone exchange

Correct answer: B

Rationale: The two-cell mechanism creates an obligatory partnership where theca cells provide androgens that granulosa cells must convert to estrogen.

MC 46. Which structural feature of the ampulla optimizes fertilization conditions?

- A. Narrow diameter for sperm concentration
- B. Absence of secretory cells
- C. Wider diameter and specialized secretory cells
- D. Lack of ciliated cells

Correct answer: C

Rationale: The ampulla's wider diameter and specialized secretory cells provide the optimal environment for fertilization.

MC 47. What triggers endometrial angiogenesis during the proliferative phase?

- A. Progesterone receptor activation
- B. Estrogen receptor activation upregulating angiogenesis genes
- C. FSH direct effects
- D. LH surge

Correct answer: B

Rationale: Estrogen receptor activation upregulates genes for angiogenesis, causing endometrial thickening during the proliferative phase.

MC 48. During follicular maturation, what indicates a follicle has become dominant?

- A. Loss of FSH receptors
- B. Reaching Graafian stage with large antrum
- C. Cessation of estrogen production
- D. Immediate ovulation

Correct answer: B

Rationale: The dominant follicle reaches Graafian stage with a large antrum filled with follicular fluid before ovulation.

MC 49. What cellular components surround the secondary oocyte at ovulation?

- A. Theca cells only
- B. Cumulus cells
- C. Endometrial cells
- D. Smooth muscle cells

Correct answer: B

Rationale: The secondary oocyte is released surrounded by cumulus cells during ovulation.

MC 50. Which hormone's surge causes enzymatic breakdown of the follicle wall?

- A. FSH
- B. Estrogen
- C. LH
- D. Progesterone

Correct answer: C

Rationale: The LH surge triggers enzymatic breakdown of the follicle wall, releasing the secondary oocyte.

MC 51. What enables sperm capacitation in the female reproductive tract?

- A. Acidic pH in the vagina
- B. Nutrients from specialized secretory cells in the ampulla
- C. Progesterone from endometrium
- D. FSH from the pituitary

Correct answer: B

Rationale: Specialized secretory cells in the ampulla produce nutrients for sperm capacitation and early embryonic development.

MC 52. In follicular development, what differentiates a secondary follicle from a primary follicle?

- A. Presence of a corpus luteum
- B. Formation of fluid-filled spaces and increased granulosa cells
- C. Complete absence of theca cells
- D. Release of the oocyte

Correct answer: B

Rationale: Secondary follicles form when granulosa cells proliferate and create fluid-filled spaces in response to FSH.

MC 53. What is the primary function of the corpus luteum?

- A. Produce estrogen for follicular development
- B. Generate FSH for the next cycle
- C. Produce progesterone to maintain the endometrium
- D. Release additional oocytes

Correct answer: C

Rationale: The corpus luteum produces progesterone that maintains the secretory endometrium.

MC 54. Which cellular mechanism prevents theca cells from producing estrogen directly?

- A. Presence of LH receptors
- B. Absence of aromatase enzyme
- C. Excessive androgen production
- D. Lack of cholesterol access

Correct answer: B

Rationale: Theca cells lack aromatase to convert androgens to estrogen despite having the complete steroidogenic pathway.

MC 55. During the proliferative phase, what endometrial change results from estrogen receptor activation?

- A. Decreased blood vessel formation
- B. Glycogen depletion
- C. DNA synthesis and cell division causing thickening
- D. Secretory protein reduction

Correct answer: C

Rationale: Estrogen receptor activation upregulates DNA synthesis and cell division genes, causing endometrial thickening.

MC 56. What coordination between cell types enables estrogen production in follicles?

- A. Granulosa cells produce androgens for theca cells
- B. Theca androgens diffuse to granulosa cells for aromatization
- C. Both cell types independently produce estrogen
- D. Oocytes directly produce estrogen

Correct answer: B

Rationale: Theca-produced androgens diffuse across the basement membrane to granulosa cells for conversion to estrogen.

MC 57. What structural adaptation of Fallopian tubes facilitates oocyte transport?

- A. Rigid walls preventing movement
- B. Combination of ciliated cells and smooth muscle peristalsis
- C. Complete absence of secretory cells
- D. Direct neural control of movement

Correct answer: B

Rationale: Ciliated epithelial cells and smooth muscle layers create coordinated movements to transport the oocyte.

MC 58. In clinical fertility treatment, why is hCG used instead of natural LH?

- A. It suppresses ovulation
- B. It mimics the LH surge for controlled timing
- C. It promotes endometrial shedding
- D. It inhibits follicular development

Correct answer: B

Rationale: hCG mimics the LH surge, allowing fertility specialists to trigger ovulation with precise timing.

MC 59. What characterizes the antrum of a Graafian follicle?

- A. Solid mass of granulosa cells
- B. Empty space with no contents
- C. Fluid containing hormones and growth factors
- D. Dense connective tissue

Correct answer: C

Rationale: The Graafian follicle's large antrum is filled with follicular fluid containing hormones and growth factors.

MC 60. Which receptor-hormone pairing drives granulosa cell proliferation?

- A. LH receptors responding to LH

- B. FSH receptors responding to FSH
- C. Estrogen receptors responding to progesterone
- D. Progesterone receptors responding to estrogen

Correct answer: B

Rationale: Granulosa cells possess FSH receptors, and FSH stimulation causes their proliferation during follicular development.

MC 61. What endometrial change supports potential embryo implantation?

- A. Rapid cell division from estrogen
- B. Decidual reaction from progesterone if implantation occurs
- C. Complete tissue breakdown
- D. Cessation of all blood flow

Correct answer: B

Rationale: Progesterone from the corpus luteum creates a decidual reaction if implantation occurs.

MC 62. During ovulation, what happens to the primary oocyte?

- A. It remains arrested in meiosis I
- B. It completes meiosis I to become a secondary oocyte
- C. It immediately completes meiosis II
- D. It undergoes mitosis

Correct answer: B

Rationale: The primary oocyte completes meiosis I to become a secondary oocyte that is released during ovulation.

MC 63. What molecular mechanism allows progesterone to prepare the endometrium for implantation?

- A. Surface receptor activation only
- B. Nuclear receptor activation shifting gene expression
- C. Direct enzyme inhibition
- D. Protein degradation

Correct answer: B

Rationale: Progesterone receptor activation acts as a transcription factor, shifting gene expression toward secretory functions and vascular maturation.

MC 64. Which cells in the follicle lack the complete steroidogenic pathway from cholesterol?

- A. Theca cells
- B. Granulosa cells
- C. Both theca and granulosa cells
- D. Neither cell type

Correct answer: B

Rationale: Granulosa cells lack the enzymatic machinery to produce androgens from cholesterol, unlike theca cells.

MC 65. What enables the female reproductive system to operate as an integrated network?

- A. Random hormone release
- B. Each structure performing specific mechanical and biochemical functions
- C. Complete independence of each organ
- D. Absence of hormonal regulation

Correct answer: B

Rationale: The system operates as an integrated network where each structure performs specific mechanical and biochemical functions essential for reproduction.

MC 66. During assisted reproduction, what does monitoring estradiol levels indicate?

- A. Corpus luteum function
- B. Follicular maturity and estrogen production
- C. Progesterone synthesis
- D. Endometrial breakdown

Correct answer: B

Rationale: Fertility specialists monitor estradiol levels to assess follicular maturity and estrogen production capacity.

MC 67. What distinguishes the secretory phase from the proliferative phase at the molecular level?

- A. FSH receptor activation versus LH receptor activation
- B. Progesterone receptor versus estrogen receptor activation
- C. Androgen versus estrogen production
- D. Mitosis versus meiosis

Correct answer: B

Rationale: The secretory phase involves progesterone receptor activation shifting gene expression, while the proliferative phase involves estrogen receptor activation.

MC 68. What happens to cervical mucus in response to follicular phase estrogen?

- A. Becomes hostile to sperm
- B. Changes to facilitate sperm transport
- C. Completely disappears
- D. Becomes highly acidic

Correct answer: B

Rationale: Increasing estrogen levels during the follicular phase cause cervical mucus changes that facilitate sperm transport.

MC 69. Which gonadotropin primarily acts on granulosa cells in the two-cell mechanism?

- A. LH
- B. FSH
- C. hCG
- D. GnRH

Correct answer: B

Rationale: Granulosa cells possess FSH receptors in the two-cell, two-gonadotropin mechanism.

MC 70. What cellular transformation characterizes corpus luteum formation?

- A. Granulosa cells dying off completely
- B. Luteinization of follicle remnants to produce progesterone
- C. New follicle development
- D. Oocyte maturation

Correct answer: B

Rationale: The follicle remnant undergoes luteinization to form the corpus luteum that produces progesterone.

MC 71. In the ampulla region, what creates the optimal fertilization environment?

- A. Narrow passages concentrating gametes
- B. Wider diameter and nutrient-producing secretory cells

- C. Absence of any cellular activity
- D. High acidity levels

Correct answer: B

Rationale: The ampulla's wider diameter and specialized secretory cells create optimal conditions for fertilization.

MC 72. What prevents granulosa cells from completing steroid hormone synthesis independently?

- A. Presence of aromatase
- B. Inability to produce androgens from cholesterol
- C. Excessive estrogen production
- D. LH receptor expression

Correct answer: B

Rationale: Granulosa cells cannot produce androgens from cholesterol, requiring theca cell cooperation.

MC 73. During IVF procedures, what allows precise oocyte retrieval timing?

- A. Random selection
- B. Correlation between follicular size and estrogen production
- C. Progesterone levels only
- D. Endometrial thickness alone

Correct answer: B

Rationale: Ultrasound tracking of follicular size correlates with estrogen production capacity for precise timing.

MC 74. What gene expression changes occur during endometrial vascular maturation?

- A. Estrogen-induced rapid cell division genes
- B. Progesterone-induced vascular development genes
- C. FSH-induced proliferation genes
- D. LH-induced breakdown genes

Correct answer: B

Rationale: Progesterone receptor activation promotes vascular maturation genes during the secretory phase.

MC 75. What structural feature distinguishes a primordial follicle from later stages?

- A. Large antrum presence
- B. Primary oocyte arrested in meiosis I with minimal granulosa cells
- C. Complete absence of any oocyte
- D. Corpus luteum formation

Correct answer: B

Rationale: Primordial follicles contain a primary oocyte arrested in meiosis I surrounded by a single layer of granulosa cells.

Unit B Review
Constructed Response Bank

Question 1

A researcher observes that when theca cells are cultured alone, they produce androgens but no estrogen, while granulosa cells cultured alone produce no steroid hormones.

- a. Explain why neither cell type can produce estrogen independently despite having different enzymatic capabilities.
- b. Describe how these two cell types cooperate through the two-cell, two-gonadotropin mechanism to produce estrogen.
- c. Identify one clinical application where understanding this cellular cooperation is essential.

Question 2

A patient undergoing fertility treatment shows multiple developing follicles on ultrasound, with the largest measuring 18mm and showing high estradiol levels.

- a. Explain the relationship between follicular size and estrogen production capacity.
- b. Describe what structural changes occur as a follicle progresses from primary to Graafian stage.
- c. State which hormone analog would be used to trigger ovulation in this patient.

Question 3

During a lecture, a student claims that the corpus luteum is just dead tissue left over after ovulation.

- a. Evaluate whether this claim is correct and explain what actually happens to the follicle after ovulation.
- b. Describe the hormonal trigger that causes this transformation and its functional significance.
- c. Name one specific endometrial change that results from corpus luteum hormone production.

Question 4

A researcher notes that Fallopian tubes have both ciliated epithelial cells and smooth muscle layers, questioning why both movement mechanisms are necessary.

- a. Explain the specific roles of ciliated epithelial cells versus smooth muscle in oocyte transport.
- b. Describe how the ampulla region's structure specifically supports fertilization.
- c. Identify which specific structure uses ciliary beating to capture the ovulated oocyte.

Question 5

Laboratory analysis shows that during the proliferative phase, endometrial cells show high expression of genes for DNA synthesis and cell division.

- a. Explain which hormone and receptor interaction causes these gene expression changes.
- b. Describe how gene expression shifts when the endometrium transitions to the secretory phase.
- c. State which ovarian structure produces the hormone responsible for secretory phase changes.

Question 6

An ultrasound technician observes that a patient's dominant follicle contains a large fluid-filled space with the oocyte attached to one side.

- a. Identify the follicular stage being observed and explain what the fluid-filled space contains.

- b. Describe the hormonal trigger that will cause this follicle to release its oocyte.
- c. State where the oocyte is initially released during ovulation.

Question 7

A fertility specialist explains that granulosa cells from small follicles produce less estrogen than those from large follicles, even with the same FSH stimulation.

- a. Explain why larger follicles have greater estrogen production capacity than smaller ones.
- b. Describe the cellular cooperation required between follicular cell types for estrogen synthesis.
- c. Identify which enzyme is unique to granulosa cells in this process.

Question 8

During ovulation, a student observes that the oocyte must travel from the ovary to the Fallopian tube despite no direct connection between these structures.

- a. Explain how the oocyte moves from the ovary to the Fallopian tube without a direct connection.
- b. Describe the two mechanisms that transport the oocyte through the Fallopian tube.
- c. Name the region of the Fallopian tube where fertilization typically occurs.

Question 9

Research data shows that women with polycystic ovary syndrome have follicles that produce androgens but reduced estrogen levels.

- a. Explain which cellular dysfunction could account for high androgens but low estrogen.
- b. Describe the normal two-cell mechanism that should convert androgens to estrogen.
- c. Identify which gonadotropin primarily stimulates granulosa cells.

Question 10

A medical student notes that endometrial cells show different gene expression patterns during proliferative versus secretory phases despite exposure to steroid hormones in both phases.

- a. Explain how different steroid hormones cause distinct gene expression patterns in the same tissue.
- b. Describe the specific gene expression changes during each phase.
- c. State which phase prepares the endometrium for potential implantation.

Question 11

An embryologist observes that during IVF, oocytes retrieved from different-sized follicles show varying maturation levels despite similar FSH treatment.

- a. Explain the relationship between follicle size and oocyte maturation status.
- b. Describe how follicular monitoring helps optimize IVF success.
- c. State what hormone analog is used to trigger final oocyte maturation.

Question 12

A researcher discovers that blocking aromatase enzyme in cultured follicles causes androgen accumulation but no estrogen production.

- a. Explain why blocking aromatase prevents estrogen production despite normal theca cell function.
- b. Describe the normal cellular cooperation that aromatase facilitates between follicular cells.
- c. Identify which cells possess aromatase enzyme in the follicle.

Question 13

Clinical observations show that women with very thin endometrium have difficulty achieving pregnancy even with normal hormone levels.

- a. Explain why endometrial thickness is critical for successful implantation.
- b. Describe the hormonal sequence that normally causes endometrial thickening and preparation.
- c. Name the specific endometrial reaction that occurs if implantation happens.

Question 14

A student questions why the ovary needs both FSH and LH receptors on different cell types rather than having all cells respond to one hormone.

- a. Explain the functional advantage of having different gonadotropin receptors on different follicular cells.
- b. Describe how this receptor distribution enables the two-cell, two-gonadotropin mechanism.
- c. State which cells possess FSH receptors.

Question 15

Laboratory tests reveal that a patient's cervical mucus remains thick throughout her cycle, potentially affecting fertility.

- a. Explain which hormone normally causes cervical mucus changes and when these changes occur.
- b. Describe how abnormal cervical mucus could affect fertility despite normal ovulation.
- c. Identify the phase when cervical mucus should be most conducive to sperm transport.

Question 16

An ultrasound shows a 22mm follicle with clear follicular fluid, but blood tests show low estradiol levels, puzzling the physician.

- a. Explain what cellular dysfunction could cause a large follicle to produce insufficient estrogen.
- b. Describe why both follicle size and hormone levels are monitored during fertility treatment.
- c. Name the enzyme that might be deficient if granulosa cells are the problem.

Question 17

Research shows that the corpus luteum maintains high progesterone production for approximately 14 days unless pregnancy occurs.

- a. Explain the transformation process that creates the corpus luteum and enables progesterone production.
- b. Describe the endometrial changes that corpus luteum progesterone induces.
- c. State what happens to the endometrium if the corpus luteum degenerates.

Question 18

A biology student wonders why meiosis I arrest in oocytes is maintained for years while sperm complete meiosis continuously.

- a. Explain the meiotic status of oocytes in primordial follicles and when this changes.
- b. Describe the functional advantage of maintaining oocytes in meiotic arrest.
- c. State which hormone surge triggers completion of meiosis I.

Question 19

During an anatomy lab, students observe that Fallopian tubes lack direct attachment to ovaries, yet oocyte transport rarely fails.

- Explain the mechanism that ensures successful oocyte capture despite the anatomical gap.
- Describe what happens to the oocyte once inside the Fallopian tube.
- Identify the initial space where the oocyte is released during ovulation.

Question 20

A researcher notes that successful IVF requires precise timing of multiple interventions based on follicular development markers.

- Explain how exogenous FSH administration affects follicular development in IVF protocols.
- Describe how monitoring helps determine optimal timing for oocyte retrieval.
- State which hormone analog triggers final oocyte maturation before retrieval.

Model Answers & Rubrics

Question 1

Part a: Explain why neither cell type can produce estrogen independently despite having different enzymatic capabilities.

Model Answer: Theca cells have the complete steroidogenic pathway to convert cholesterol to androgens and possess LH receptors, but they lack the aromatase enzyme needed to convert androgens to estrogen. Granulosa cells possess aromatase enzyme and FSH receptors but lack the enzymatic machinery to produce androgens from cholesterol. This creates an obligatory partnership where neither cell type can complete estrogen synthesis alone. (2 marks) (2 marks)

Part b: Describe how these two cell types cooperate through the two-cell, two-gonadotropin mechanism to produce estrogen.

Model Answer: In the two-cell, two-gonadotropin mechanism, LH stimulates theca cells to produce androgens from cholesterol. These androgens diffuse across the basement membrane to reach granulosa cells. FSH stimulates granulosa cells, which use their aromatase enzyme to convert the theca-derived androgens into estrogen. This cooperation enables complete estrogen synthesis. (2 marks) (2 marks)

Part c: Identify one clinical application where understanding this cellular cooperation is essential.

Model Answer: In controlled ovarian stimulation for IVF, fertility specialists use exogenous FSH to stimulate granulosa cells and monitor estradiol levels to assess follicular maturity, knowing that proper estrogen production requires both cell types working together. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining theca cells lack aromatase, 1 mark for explaining granulosa cells lack androgen synthesis machinery

Part b (2 marks): 1 mark for describing androgen diffusion from theca to granulosa, 1 mark for explaining gonadotropin roles (LH on theca, FSH on granulosa)

Part c (1 mark): 1 mark for identifying a relevant clinical application such as IVF monitoring

Question 2

Part a: Explain the relationship between follicular size and estrogen production capacity.

Model Answer: As follicles mature and increase in size, they contain more granulosa cells that proliferate in response to FSH stimulation. These increased granulosa cells have greater aromatase enzyme capacity to convert theca-derived androgens to estrogen. The larger follicle size correlates with higher estrogen production capacity, which is why follicular size tracking helps assess maturity. (2 marks) (2 marks)

Part b: Describe what structural changes occur as a follicle progresses from primary to Graafian stage.

Model Answer: A primary follicle has a single layer of granulosa cells around the oocyte. FSH stimulation causes granulosa cell proliferation and formation of fluid-filled spaces, creating a secondary follicle. The dominant follicle continues developing into a Graafian follicle with a large antrum filled with follicular fluid containing hormones and growth factors, ready for ovulation. (2 marks) (2 marks)

Part c: State which hormone analog would be used to trigger ovulation in this patient.

Model Answer: hCG (human chorionic gonadotropin) would be used because it mimics the LH surge to trigger ovulation with precise timing. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining granulosa cell proliferation with size, 1 mark for connecting to increased aromatase/estrogen capacity

Part b (2 marks): 1 mark for describing early follicular changes, 1 mark for describing Graafian characteristics

Part c (1 mark): 1 mark for correctly identifying hCG

Question 3

Part a: Evaluate whether this claim is correct and explain what actually happens to the follicle after ovulation.

Model Answer: The claim is incorrect. After ovulation, the follicle remnant undergoes luteinization, transforming into the corpus luteum, which is metabolically active tissue. The corpus luteum produces progesterone essential for maintaining the secretory endometrium and supporting potential implantation, not dead tissue. (2 marks) (2 marks)

Part b: Describe the hormonal trigger that causes this transformation and its functional significance.

Model Answer: The LH surge that triggers ovulation also causes luteinization of the follicle remnant. This transformation is functionally significant because the resulting corpus luteum produces progesterone that maintains the secretory endometrium and creates a decidual reaction if implantation occurs, essential for early pregnancy support. (2 marks) (2 marks)

Part c: Name one specific endometrial change that results from corpus luteum hormone production.

Model Answer: The corpus luteum's progesterone production causes glycogen accumulation in endometrial cells to support potential embryo implantation. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for stating claim is incorrect, 1 mark for explaining luteinization and progesterone production

Part b (2 marks): 1 mark for identifying LH surge as trigger, 1 mark for explaining functional significance

Part c (1 mark): 1 mark for identifying a specific change like glycogen accumulation or vascular maturation

Question 4

Part a: Explain the specific roles of ciliated epithelial cells versus smooth muscle in oocyte transport.

Model Answer: Ciliated epithelial cells create coordinated beating movements that help capture and transport the oocyte, particularly important at the fimbriae for initial oocyte capture. Smooth muscle layers create peristaltic contractions that provide the main force for moving the oocyte through the tube toward the uterus. Both work together for effective transport. (2 marks) (2 marks)

Part b: Describe how the ampulla region's structure specifically supports fertilization.

Model Answer: The ampulla has a wider diameter that slows oocyte transport, providing more time for sperm-egg interaction. It contains specialized secretory cells that produce nutrients supporting sperm capacitation and early embryonic development, creating an optimal environment for fertilization to occur. (2 marks) (2 marks)

Part c: Identify which specific structure uses ciliary beating to capture the ovulated oocyte.

Model Answer: The fimbriae at the tube opening use coordinated ciliary beating to capture the released ovum from the peritoneal cavity. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining ciliated cell function, 1 mark for explaining smooth muscle function

Part b (2 marks): 1 mark for describing wider diameter significance, 1 mark for explaining secretory cell function

Part c (1 mark): 1 mark for correctly identifying fimbriae

Question 5

Part a: Explain which hormone and receptor interaction causes these gene expression changes.

Model Answer: Estrogen binds to nuclear estrogen receptors in endometrial cells. These activated estrogen receptors act as transcription factors that upregulate genes for DNA synthesis, cell division, and angiogenesis, causing the observed gene expression pattern during the proliferative phase. (2 marks) (2 marks)

Part b: Describe how gene expression shifts when the endometrium transitions to the secretory phase.

Model Answer: During the secretory phase, progesterone receptor activation shifts gene expression away from proliferation toward secretory protein production, glycogen accumulation, and vascular maturation genes. This prepares the endometrium to support potential embryo implantation rather than continued growth. (2 marks) (2 marks)

Part c: State which ovarian structure produces the hormone responsible for secretory phase changes.

Model Answer: The corpus luteum produces progesterone responsible for secretory phase endometrial changes. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying estrogen and its receptor, 1 mark for explaining transcription factor mechanism

Part b (2 marks): 1 mark for describing shift from proliferation genes, 1 mark for identifying secretory/supportive genes

Part c (1 mark): 1 mark for correctly identifying corpus luteum

Question 6

Part a: Identify the follicular stage being observed and explain what the fluid-filled space contains.

Model Answer: This is a Graafian (mature) follicle. The large fluid-filled space is the antrum, which contains follicular fluid rich in hormones (especially estrogen) and growth factors that support oocyte development and will be released during ovulation. (2 marks) (2 marks)

Part b: Describe the hormonal trigger that will cause this follicle to release its oocyte.

Model Answer: The LH surge, triggered by peak estrogen levels from the mature follicle, will cause enzymatic breakdown of the follicle wall. This releases the secondary oocyte surrounded by cumulus cells into the peritoneal cavity during ovulation. (2 marks) (2 marks)

Part c: State where the oocyte is initially released during ovulation.

Model Answer: The oocyte is initially released into the peritoneal cavity before being captured by the fimbriae. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying Graafian follicle, 1 mark for describing antrum contents

Part b (2 marks): 1 mark for identifying LH surge, 1 mark for describing enzymatic breakdown mechanism

Part c (1 mark): 1 mark for correctly identifying peritoneal cavity

Question 7

Part a: Explain why larger follicles have greater estrogen production capacity than smaller ones.

Model Answer: Larger follicles contain more granulosa cells due to FSH-stimulated proliferation over time. With more granulosa cells present, there is greater total aromatase enzyme capacity to convert theca-derived androgens to estrogen, resulting in higher estrogen production despite the same FSH stimulation level. (2 marks) (2 marks)

Part b: Describe the cellular cooperation required between follicular cell types for estrogen synthesis.

Model Answer: Theca cells respond to LH and produce androgens from cholesterol using their complete steroidogenic pathway. These androgens diffuse across the basement membrane to granulosa cells, which respond to FSH and use aromatase to convert the androgens to estrogen, completing the two-cell mechanism. (2 marks) (2 marks)

Part c: Identify which enzyme is unique to granulosa cells in this process.

Model Answer: Aromatase enzyme is unique to granulosa cells and essential for converting androgens to estrogen. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining increased cell number, 1 mark for connecting to greater enzyme capacity

Part b (2 marks): 1 mark for describing theca cell androgen production, 1 mark for describing granulosa aromatization

Part c (1 mark): 1 mark for correctly identifying aromatase

Question 8

Part a: Explain how the oocyte moves from the ovary to the Fallopian tube without a direct connection.

Model Answer: The oocyte is released into the peritoneal cavity during ovulation. The fimbriae at the opening of the Fallopian tube use coordinated ciliary beating to create currents that capture the oocyte from the peritoneal cavity and sweep it into the tube. (2 marks) (2 marks)

Part b: Describe the two mechanisms that transport the oocyte through the Fallopian tube.

Model Answer: Ciliated epithelial cells lining the tube create coordinated beating movements that help move the oocyte along. Smooth muscle layers in the tube wall create peristaltic contractions that provide the main propulsive force to transport the oocyte toward the uterus. (2 marks) (2 marks)

Part c: Name the region of the Fallopian tube where fertilization typically occurs.

Model Answer: Fertilization typically occurs in the ampulla region of the Fallopian tube. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for mentioning peritoneal cavity release, 1 mark for explaining fimbrial capture

Part b (2 marks): 1 mark for describing ciliary action, 1 mark for describing peristaltic contractions

Part c (1 mark): 1 mark for correctly identifying ampulla

Question 9

Part a: Explain which cellular dysfunction could account for high androgens but low estrogen.

Model Answer: The dysfunction likely involves granulosa cells having reduced aromatase enzyme activity or expression. While theca cells continue producing androgens normally, the granulosa cells cannot efficiently convert these androgens to estrogen, leading to androgen accumulation and reduced estrogen production. (2 marks) (2 marks)

Part b: Describe the normal two-cell mechanism that should convert androgens to estrogen.

Model Answer: Normally, theca cells produce androgens that diffuse to granulosa cells. Granulosa cells use aromatase enzyme to convert these androgens to estrogen. This two-cell, two-gonadotropin mechanism requires both LH stimulation of theca cells and FSH stimulation of granulosa cells for complete estrogen synthesis. (2 marks) (2 marks)

Part c: Identify which gonadotropin primarily stimulates granulosa cells.

Model Answer: FSH (follicle-stimulating hormone) primarily stimulates granulosa cells. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying granulosa/aromatase dysfunction, 1 mark for explaining the consequence

Part b (2 marks): 1 mark for describing androgen diffusion, 1 mark for explaining aromatase conversion

Part c (1 mark): 1 mark for correctly identifying FSH

Question 10

Part a: Explain how different steroid hormones cause distinct gene expression patterns in the same tissue.

Model Answer: Estrogen and progesterone bind to different nuclear receptors (estrogen receptors vs progesterone receptors). Each activated receptor complex acts as a specific transcription factor that regulates different sets of genes, causing distinct expression patterns despite both being steroid hormones acting on the same endometrial tissue. (2 marks) (2 marks)

Part b: Describe the specific gene expression changes during each phase.

Model Answer: During the proliferative phase, estrogen receptor activation upregulates genes for DNA synthesis, cell division, and angiogenesis, causing endometrial thickening. During the secretory phase, progesterone receptor activation shifts expression toward secretory protein production, glycogen accumulation, and vascular maturation genes. (2 marks) (2 marks)

Part c: State which phase prepares the endometrium for potential implantation.

Model Answer: The secretory phase prepares the endometrium for potential implantation. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining different receptor binding, 1 mark for transcription factor concept

Part b (2 marks): 1 mark for proliferative phase genes, 1 mark for secretory phase genes

Part c (1 mark): 1 mark for correctly identifying secretory phase

Question 11

Part a: Explain the relationship between follicle size and oocyte maturation status.

Model Answer: Larger follicles have undergone more development time with greater granulosa cell proliferation and hormone production. This extended development allows the oocyte more time to progress through maturation, including completion of meiosis I to become a secondary oocyte ready for fertilization, compared to smaller follicles. (2 marks) (2 marks)

Part b: Describe how follicular monitoring helps optimize IVF success.

Model Answer: Ultrasound tracking of follicular size correlates with estrogen production capacity and oocyte maturity. Monitoring estradiol levels confirms follicular function. This allows precise timing of hCG administration to trigger ovulation when multiple follicles reach optimal maturity, maximizing the number of competent oocytes retrieved. (2 marks) (2 marks)

Part c: State what hormone analog is used to trigger final oocyte maturation.

Model Answer: hCG (human chorionic gonadotropin) is used to mimic the LH surge. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining size-development relationship, 1 mark for connecting to oocyte maturation

Part b (2 marks): 1 mark for describing monitoring methods, 1 mark for explaining timing optimization

Part c (1 mark): 1 mark for correctly identifying hCG

Question 12

Part a: Explain why blocking aromatase prevents estrogen production despite normal theca cell function.

Model Answer: Aromatase is the essential enzyme that converts androgens to estrogen, found only in granulosa cells. When blocked, theca cells continue producing androgens normally, but granulosa cells cannot convert these androgens to estrogen, causing androgen accumulation without estrogen synthesis. (2 marks) (2 marks)

Part b: Describe the normal cellular cooperation that aromatase facilitates between follicular cells.

Model Answer: In normal function, theca cells produce androgens that diffuse across the basement membrane to granulosa cells. Granulosa cells use aromatase to convert these theca-derived androgens into estrogen. This two-cell mechanism creates an obligatory partnership where both cell types are essential for estrogen production. (2 marks) (2 marks)

Part c: Identify which cells possess aromatase enzyme in the follicle.

Model Answer: Granulosa cells possess aromatase enzyme. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining aromatase function, 1 mark for describing the consequence of blocking

Part b (2 marks): 1 mark for describing androgen diffusion, 1 mark for explaining the partnership requirement

Part c (1 mark): 1 mark for correctly identifying granulosa cells

Question 13

Part a: Explain why endometrial thickness is critical for successful implantation.

Model Answer: Adequate endometrial thickness provides sufficient secretory glands and glycogen stores to nourish the early embryo. It also ensures proper vascular development for blood supply and creates the appropriate cellular environment for the decidual reaction necessary to support implantation and early pregnancy. (2 marks) (2 marks)

Part b: Describe the hormonal sequence that normally causes endometrial thickening and preparation.

Model Answer: During the proliferative phase, estrogen from developing follicles activates endometrial estrogen receptors, upregulating genes for DNA synthesis and cell division, causing thickening. After ovulation, progesterone from the corpus luteum shifts the endometrium to secretory phase with glycogen accumulation and vascular maturation. (2 marks) (2 marks)

Part c: Name the specific endometrial reaction that occurs if implantation happens.

Model Answer: The decidual reaction occurs if implantation happens. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining nutritive function, 1 mark for describing structural support

Part b (2 marks): 1 mark for proliferative phase changes, 1 mark for secretory phase changes

Part c (1 mark): 1 mark for correctly identifying decidual reaction

Question 14

Part a: Explain the functional advantage of having different gonadotropin receptors on different follicular cells.

Model Answer: Having LH receptors on theca cells and FSH receptors on granulosa cells allows precise control of the two-step estrogen synthesis pathway. This separation enables independent regulation of androgen production (via LH) and aromatization to estrogen (via FSH), providing fine-tuned hormonal control of follicular development. (2 marks) (2 marks)

Part b: Describe how this receptor distribution enables the two-cell, two-gonadotropin mechanism.

Model Answer: LH stimulates theca cells to produce androgens from cholesterol. These androgens diffuse to granulosa cells, which are stimulated by FSH to express aromatase and convert the androgens to estrogen. This creates an obligatory partnership where both cell types and both gonadotropins are required for estrogen synthesis. (2 marks) (2 marks)

Part c: State which cells possess FSH receptors.

Model Answer: Granulosa cells possess FSH receptors. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining independent regulation, 1 mark for describing control advantage

Part b (2 marks): 1 mark for describing LH-theca interaction, 1 mark for describing FSH-granulosa interaction

Part c (1 mark): 1 mark for correctly identifying granulosa cells

Question 15

Part a: Explain which hormone normally causes cervical mucus changes and when these changes occur.

Model Answer: Estrogen from developing follicles during the follicular phase causes cervical mucus changes. Rising estrogen levels make the mucus thinner and more alkaline, facilitating sperm transport through the cervix toward the oocyte for potential fertilization. (2 marks) (2 marks)

Part b: Describe how abnormal cervical mucus could affect fertility despite normal ovulation.

Model Answer: Thick cervical mucus creates a barrier that prevents sperm from passing through the cervix to reach the Fallopian tubes. Even with normal ovulation releasing a healthy oocyte, fertilization cannot occur if sperm cannot traverse the cervical barrier to reach the oocyte in the ampulla. (2 marks) (2 marks)

Part c: Identify the phase when cervical mucus should be most conducive to sperm transport.

Model Answer: The follicular phase, particularly near ovulation when estrogen peaks. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying estrogen, 1 mark for describing mucus changes

Part b (2 marks): 1 mark for explaining barrier effect, 1 mark for connecting to fertilization failure

Part c (1 mark): 1 mark for identifying follicular phase/near ovulation

Question 16

Part a: Explain what cellular dysfunction could cause a large follicle to produce insufficient estrogen.

Model Answer: The dysfunction likely involves either inadequate granulosa cell function (reduced aromatase activity or cell number) or impaired theca cell androgen production. Without proper function of either cell type in the two-cell mechanism, estrogen production is compromised despite the follicle reaching normal size. (2 marks) (2 marks)

Part b: Describe why both follicle size and hormone levels are monitored during fertility treatment.

Model Answer: Follicle size indicates physical maturity and potential oocyte development, while estradiol levels confirm functional hormone production by the follicular cells. Monitoring both ensures the follicle is not just structurally mature but also functionally competent, as size alone doesn't guarantee proper hormone production. (2 marks) (2 marks)

Part c: Name the enzyme that might be deficient if granulosa cells are the problem.

Model Answer: Aromatase enzyme might be deficient in the granulosa cells. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying cellular dysfunction, 1 mark for explaining the mechanism

Part b (2 marks): 1 mark for explaining size monitoring, 1 mark for explaining hormone monitoring importance

Part c (1 mark): 1 mark for correctly identifying aromatase

Question 17

Part a: Explain the transformation process that creates the corpus luteum and enables progesterone production.

Model Answer: After ovulation triggered by the LH surge, the follicle remnant undergoes luteinization. The granulosa and theca cells transform into luteal cells that acquire the ability to produce large amounts of progesterone, creating the corpus luteum structure specialized for progesterone synthesis. (2 marks) (2 marks)

Part b: Describe the endometrial changes that corpus luteum progesterone induces.

Model Answer: Progesterone from the corpus luteum maintains the secretory endometrium by shifting gene expression toward secretory protein production, glycogen accumulation for embryo nutrition, and vascular maturation. If implantation occurs, it triggers a decidual reaction to support the pregnancy. (2 marks) (2 marks)

Part c: State what happens to the endometrium if the corpus luteum degenerates.

Model Answer: The endometrium breaks down and is shed during menstruation. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for describing luteinization, 1 mark for explaining progesterone production

Part b (2 marks): 1 mark for secretory changes, 1 mark for implantation support

Part c (1 mark): 1 mark for identifying menstruation/shedding

Question 18

Part a: Explain the meiotic status of oocytes in primordial follicles and when this changes.

Model Answer: Primary oocytes in primordial follicles are arrested in prophase of meiosis I. This arrest is maintained until the LH surge triggers ovulation, when the primary oocyte completes meiosis I to become a secondary oocyte, which is then arrested in meiosis II until fertilization. (2 marks) (2 marks)

Part b: Describe the functional advantage of maintaining oocytes in meiotic arrest.

Model Answer: Meiotic arrest preserves oocytes in a metabolically quiet state, protecting them from damage over decades of storage. It also ensures oocytes only complete development when hormonal conditions indicate optimal timing for fertilization, coordinating gamete maturation with the reproductive cycle. (2 marks) (2 marks)

Part c: State which hormone surge triggers completion of meiosis I.

Model Answer: The LH surge triggers completion of meiosis I. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for describing meiosis I arrest, 1 mark for explaining when it changes

Part b (2 marks): 1 mark for preservation benefit, 1 mark for timing coordination

Part c (1 mark): 1 mark for correctly identifying LH surge

Question 19

Part a: Explain the mechanism that ensures successful oocyte capture despite the anatomical gap.

Model Answer: The fimbriae at the Fallopian tube opening position near the ovary and use coordinated ciliary beating to create fluid currents. These currents capture the oocyte released into the peritoneal cavity during ovulation and sweep it into the tube, bridging the anatomical gap effectively. (2 marks) (2 marks)

Part b: Describe what happens to the oocyte once inside the Fallopian tube.

Model Answer: Once captured, the oocyte is transported through the tube by two mechanisms: ciliated epithelial cells create directional flow, while smooth muscle layers produce peristaltic contractions. The oocyte moves through the ampulla where fertilization may occur, then continues toward the uterus. (2 marks) (2 marks)

Part c: Identify the initial space where the oocyte is released during ovulation.

Model Answer: The oocyte is initially released into the peritoneal cavity. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for describing fimbrial function, 1 mark for explaining capture mechanism

Part b (2 marks): 1 mark for transport mechanisms, 1 mark for describing pathway

Part c (1 mark): 1 mark for correctly identifying peritoneal cavity

Question 20

Part a: Explain how exogenous FSH administration affects follicular development in IVF protocols.

Model Answer: Exogenous FSH stimulates multiple follicles to develop simultaneously by promoting granulosa cell proliferation and estrogen production in each follicle. This contrasts with natural cycles where typically only one follicle becomes dominant, allowing retrieval of multiple mature oocytes for fertilization attempts. (2 marks) (2 marks)

Part b: Describe how monitoring helps determine optimal timing for oocyte retrieval.

Model Answer: Ultrasound tracking shows follicular size, which correlates with maturity, while estradiol level monitoring confirms functional estrogen production capacity. When multiple follicles reach 18-20mm with appropriate estradiol levels, hCG is administered to trigger final maturation, with retrieval timed 34-36 hours later. (2 marks) (2 marks)

Part c: State which hormone analog triggers final oocyte maturation before retrieval.

Model Answer: hCG (human chorionic gonadotropin) triggers final maturation. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining multiple follicle stimulation, 1 mark for contrasting with natural cycle

Part b (2 marks): 1 mark for monitoring methods, 1 mark for timing determination

Part c (1 mark): 1 mark for correctly identifying hCG